Chapter 2, Problem 69P

Problem

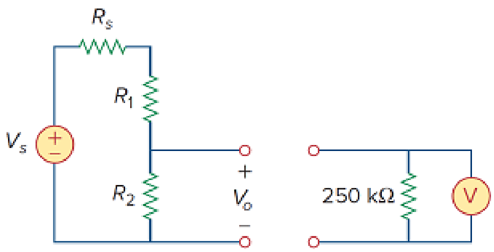
A voltmeter is used to measure *Vo* in the circuit in Fig. 2.129. The voltmeter model consists of an ideal voltmeter in parallel with a 250-kΩ resistor. Let *Vs* = 95 V, *Rs* = 25 kΩ, and *R*1 = 40 kΩ. Calculate *Vo* with and without the voltmeter when

(a) *R*2 = 5 kΩ

(b) *R*2 = 25 kΩ

(c) *R*2 = 250 kΩ

Figure 2.129:



Step-by-step solution

Step 1 of 8

Refer to Figure 2.129 in the textbook.

Draw the circuit diagram by combining both the circuits.

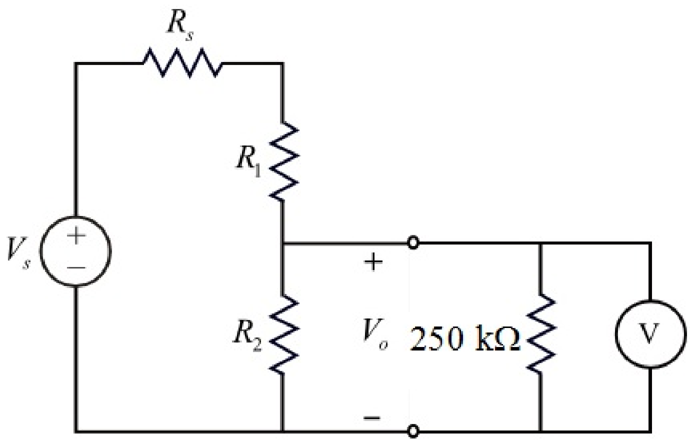


Figure 1

From the Figure 1, calculate the voltage V_o using voltage divider rule.

$$V_o = (V_s)\left(\frac{(R_2 \,||\, R_m)}{R_s + R_1 + (R_2 \,||\, R_m)}\right) \quad \ldots\ldots (1)$$

Step 2 of 8

Consider the following circuit diagram without voltmeter.

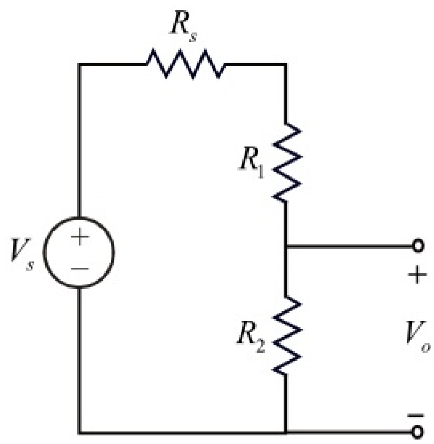


Figure 2

From the Figure 2, calculate the voltage V_o using voltage divider rule.

$$V_o = (V_s)\left(\frac{R_2}{R_s + R_1 + R_2}\right) \quad \text{......} (2)$$

Step 3 of 8

(a)

Consider the following data:

The value of resistor, $R_2 = 5 \text{ k}\Omega$

Calculate the voltage V_o with the voltmeter when $R_2 = 5 \text{ k}\Omega$ using equation (1).

$$\begin{aligned}
V_o &= (95)\left(\frac{(5\text{k} \parallel 250\text{k})}{25\text{k} + 40\text{k} + (5\text{k} \parallel 250\text{k})}\right) \\
&= (95)\left(\frac{\frac{(5\text{k})(250\text{k})}{5\text{k} + 250\text{k}}}{65\text{k} + \left(\frac{(5\text{k})(250\text{k})}{5\text{k} + 250\text{k}}\right)}\right) \\
&= (95)\left(\frac{4.9\text{k}}{65\text{k} + 4.9\text{k}}\right) \\
&= 6.66 \text{ V}
\end{aligned}$$

Therefore, the voltage V_o with the voltmeter when $R_2 = 1 \text{ k}\Omega$ is $\boxed{6.66 \text{ V}}$.

Step 4 of 8

Calculate the voltage V_o without voltmeter when $R_2 = 5 \text{ k}\Omega$ using equation (2).

$$\begin{aligned}
V_o &= (95)\left(\frac{5\text{k}}{25\text{k} + 40\text{k} + 5\text{k}}\right) \\
&= (95)\left(\frac{5\text{k}}{70\text{k}}\right) \\
&= 6.786 \text{ V}
\end{aligned}$$

Therefore, the voltage V_o without voltmeter when $R_2 = 5 \text{ k}\Omega$ is $\boxed{6.786 \text{ V}}$.

Step 5 of 8

(b)

Calculate the voltage V_o with the voltmeter when $R_2 = 25\text{ k}\Omega$ using equation (1).

$$\begin{aligned}
V_o &= (95)\left(\frac{(25\text{k} \parallel 250\text{k})}{25\text{k} + 40\text{k} + (25\text{k} \parallel 250\text{k})}\right) \\
&= (95)\left(\frac{\dfrac{(25\text{k})(250\text{k})}{25\text{k}+250\text{k}}}{65\text{k} + \left(\dfrac{(25\text{k})(250\text{k})}{25\text{k}+250\text{k}}\right)}\right) \\
&= (95)\left(\frac{22.73\text{k}}{65\text{k}+22.73\text{k}}\right) \\
&= 24.61\text{ V}
\end{aligned}$$

Therefore, the voltage V_o with the voltmeter when $R_2 = 25\text{ k}\Omega$ is $\boxed{24.61\text{ V}}$.

Step 6 of 8

Calculate the voltage V_o without voltmeter when $R_2 = 25\text{ k}\Omega$ using equation (2).

$$\begin{aligned}
V_o &= (95)\left(\frac{25\text{k}}{25\text{k}+40\text{k}+25\text{k}}\right) \\
&= (95)\left(\frac{25\text{k}}{90\text{k}}\right) \\
&= 26.39\text{ V}
\end{aligned}$$

Therefore, the voltage V_o without voltmeter when $R_2 = 25\text{ k}\Omega$ is $\boxed{26.39\text{ V}}$.

Step 7 of 8

(c)

Calculate the voltage V_o with the voltmeter when $R_2 = 250\text{ k}\Omega$ using equation (1).

$$\begin{aligned}
V_o &= (95)\left(\frac{(250\text{k} \parallel 250\text{k})}{25\text{k} + 40\text{k} + (250\text{k} \parallel 250\text{k})}\right) \\
&= (95)\left(\frac{\dfrac{(250\text{k})(250\text{k})}{250\text{k}+250\text{k}}}{65\text{k} + \left(\dfrac{(250\text{k})(250\text{k})}{250\text{k}+250\text{k}}\right)}\right) \\
&= (95)\left(\frac{125\text{k}}{65\text{k}+125\text{k}}\right) \\
&= 62.5\text{ V}
\end{aligned}$$

Therefore, the voltage V_o with the voltmeter when $R_2 = 250\text{ k}\Omega$ is $\boxed{62.5\text{ V}}$.

Step 8 of 8

Calculate the voltage V_o without voltmeter when $R_2 = 250\text{ k}\Omega$ using equation (2).

$$\begin{aligned}
V_o &= (95)\left(\frac{250\text{k}}{25\text{k}+40\text{k}+250\text{k}}\right) \\
&= (95)\left(\frac{250\text{k}}{315\text{k}}\right) \\
&= 75.4\text{ V}
\end{aligned}$$

Therefore, the voltage V_o without voltmeter when $R_2 = 250\text{ k}\Omega$ is $\boxed{75.4\text{ V}}$.